

10. Least Recent Use Page Replacement

```
# include <stdio.h>
```

```
int main () {
    int n,i,j,k,frames,faults=0;

    printf("Enter the number of pages: ");
    scanf("%d",&n);

    int pages[n];
    printf("Enter page reference string: ");
    for (i=0;i<n;i++) {
        scanf("%d",&pages[i]);
    }

    printf("Enter the number of frames: ");
    scanf("%d",&frames);

    int frame[frames],time[frames];

    for (i=0;i<frames;i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    int counter = 0;

    for (i=0;i<n;i++) {
        int found = 0;

        for (j=0;j<frames;j++) {
            if (frame[j] == pages[i]) {
                counter++;
                found = 1;
                time[j] = counter;
                break;
            }
        }

        if (!found) {
            int min = time[0];
            k = 0;

            for (j=1;j<frames;j++) {
                if (time[j] < min) {
                    min = time[j];
                    k = j;
                }
            }
        }
    }
}
```

```
        }
        frame[k] = pages[i];
        counter++;
        time[k] = counter;
        faults++;
    }
}

printf("Total page faults: %d", faults);

return 0;
}
```